Physics for Engineers

Formula booklet

Electricity Part 1: foundations of DC circuit theory · Thermodynamics Part 1: description of a system · heat engines

How to read this booklet

Each row gives a quantity and where it is in the notes (Box number, equation or page), the formula, then the symbols with their units and when the formula applies. Page numbers refer to the lecture notes, whose blank boxes are filled in here.

- † marks a result that is **not printed in the notes**: a direct consequence, a standard result used in the exercises, or the material of Part III. Use it when it helps, but quote the notes' form when there is one.
- Part III (work, heat, heat engines) goes beyond Part 1 of the Thermodynamics notes. It includes the heat taken in by an isothermal expansion, $Q = nRT_H \ln(V_{\max}/V_{\min})$.

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Constants

Elementary charge	$e = 1.6 \times 10^{-19}$ C
Avogadro constant	$N_A = 6.022 \times 10^{23}$ mol^{-1}
Ideal gas constant	$R = 8.314$ J mol^{-1} K^{-1} (≈ 8.3)
Boltzmann constant	$k_B = R/N_A = 1.38 \times 10^{-23}$ J K^{-1}†
Gravity	$g = 9.81$ m s^{-2}
At 300 K	$RT \approx 2.5 \times 10^3$ J mol^{-1}

Conversions

$T\,(\mathrm{K}) = \theta\,(^\circ\mathrm{C}) + 273.15$

1 bar $= 10^5$ Pa; 1 atm $= 1.013 \times 10^5$ Pa

1 L $= 1$ dm$^3 = 10^{-3}$ m^3; 1 mL $= 1$ cm$^3 = 10^{-6}$ m^3

1 mm$^2 = 10^{-6}$ m^2; 1 cm$^2 = 10^{-4}$ m^2

1 W h $= 3600$ J; 1 kW h $= 3.6 \times 10^6$ J

1 A h $= 3600$ C; 1 kΩ $\times$ 1 mA $= 1$ V

SI prefixes

G (giga) 10^9	M (mega) 10^6	k (kilo) 10^3
m (milli) 10^{-3}	µ (micro) 10^{-6}	n (nano) 10^{-9}
p (pico) 10^{-12}		

Exact symbols: k (kilo) ≠ K (kelvin), m (milli) ≠ M (mega).

Part I
Electricity: foundations of DC circuit theory

Notes Elec-S1 Part 1 (38 pages). Arrows: the arrow of u_{AB} points towards A. Receiver convention: u and i arrows opposite; generator convention: same direction.

1.A Electric voltage (pp. 2–7)

Electric voltage Box 1	$u_{AB} = V_A - V_B$	Volt (V). Only potential differences can be measured; potential and voltage have the same unit.
Sign of a voltage eq. (1)	$u_{AB} = -u_{BA}$	$u_{AB} > 0 \Leftrightarrow V_A > V_B$.
Adding voltages†	$u_{AC} = u_{AB} + u_{BC}$	Potentials subtract like heights; the basis of the mesh law.
Ground p. 2	$V_{\text{ground}} = 0$	Reference of potentials: the wire on the − terminal of the source. "Earth" if it is connected to the Earth.
Ideal voltage source Box 2	$u = e \quad$ whatever i	e: electromotive force, in volts (not newtons). A real source has an internal resistance: u drops as i grows (see the Thévenin model, 3.B).
Voltmeter p. 7	$R_V \to \infty$	In parallel, COM on the − side (otherwise the sign flips); $R_V = 1$ to 10 MΩ.

Orders of magnitude (Tab. 1): neuron ∼100 mV; AA cell 1.5 V; Li-ion 3.6–4.2 V; mains 230 V_{rms}; high-speed train line 25 kV; extra-high-voltage line 400 kV.

1.B Electric current (pp. 7–11)

Electric current Box 3	$i = \dfrac{\mathrm{d}q}{\mathrm{d}t}$	Ampere (A), an SI base unit: 1 A = 1 C s^{-1}. Same current all along a series path. Conventional direction: that of positive charges.
Charge transported†	$q = \displaystyle\int_{t_1}^{t_2} i\,\mathrm{d}t \qquad \text{DC: } q = i\,\Delta t$	Coulomb (C).
Number of electrons†	$N = \dfrac{\lvert q\rvert}{e} = \dfrac{i\,\Delta t}{e}$	$e = 1.6 \times 10^{-19}$ C.
Microscopic expression Box 4, eq. (2)	$i = n\,q\,v\,S$	n carrier density (m^{-3}), q carrier charge (C), v drift velocity (m s^{-1}), S cross-section (m^2). Electrons: $i = -nevS$.
Drift velocity†	$v = \dfrac{i}{n\,e\,S}$	Copper, 1 mm^2, 1 A: $v \approx 7 \times 10^{-5}$ m s^{-1}. Not the speed of the signal.
Current density†	$j = \dfrac{i}{S} = n\,\lvert q\rvert\,v$	A m^{-2}.
Ammeter p. 11	$R_A \to 0$	In series (open the branch to insert it); $R_A < 1\ \Omega$.

1.C Circuits and two-terminal components (pp. 11–14)

Circuit vocabulary Box 5	node: junction of ≥ 3 conductors branch: between two nodes	Loop: any closed path. Mesh: a loop containing no other loop. Points joined only by wire form one (electrically distinct) node.
Counting equations 3.A	$N - 1$ node laws, $\quad M = B - N + 1$ meshes	N distinct nodes, B branches: B equations for the B branch currents.
Series and parallel Box 6	series: same i, $\quad u = \sum_k u_k$ parallel: same u, $\quad i = \sum_k i_k$	Same current in series (node law), voltages add (mesh law). Same voltage in parallel (mesh law), currents add at the nodes (node law).
Direction of the current p. 13	outside the source: from ⊕ to ⊖	Electrons flow the other way ($i = -nevS$).
Electrical power Box 7	$p = u\,i$	Watt: 1 W = 1 V × 1 A. Receiver convention: $p > 0$ is received; generator convention: $p > 0$ is supplied.

Electrical energy eqs. (4), (5)	$E = \int_{t_1}^{t_2} u\, i \,\mathrm{d}t \qquad \text{DC: } E = u_0\, i_0\, \Delta t$	Joule (J); 1 W h = 3600 J.

2.A Characteristic curve (pp. 14–19)

Characteristic Box 8	$i = f(u) \quad \text{or} \quad u = g(i)$	Pairs (u_k, i_k) measured in many states. Depends on the convention: receiver for a passive component, generator for an active one.
Resistor Box 9	$i = \dfrac{u}{R}$: a line through O	Slope of $i = f(u)$: $G = 1/R$. Slope of $u = g(i)$: R. The signature of an ohmic conductor.
Quick (naive) estimate of R Box 10	$R = \dfrac{1}{N}\sum_{k=1}^{N} \dfrac{u_k}{i_k}$	Mean of the ratios $r_k = u_k/i_k$; treats the noisy low-current points like the others.
Estimate by linear regression Box 11	$\hat{R} = \dfrac{\sum_k u_k\, i_k}{\sum_k i_k^2}$	Minimises $\phi(R) = \sum_k (u_k - R\, i_k)^2$: best line $u = \hat{R}\, i$ through O.
Standard uncertainty on $\hat{R}$ p. 18	$u(\hat{R}) = \sqrt{\dfrac{\frac{1}{N-1}\sum_k (u_k - \hat{R}\, i_k)^2}{\sum_k i_k^2}}$	$u_k - \hat{R}\, i_k$: residuals (Tab. 4).
Other characteristics Figs. 15, 19	not lines through O	Ideal voltage source: vertical line $u = E$. Ideal current source: horizontal line $i = I_0$. Lamp: curved. Diode, Zener diode ($-U_Z$).

2.B Ohm's law and Joule's law (pp. 19–21)

Ohm's law Box 12	$u = R\, i$	Receiver convention ($u = -R\, i$ in generator convention). R in ohms (Ω).
Conductance eq. (6)	$G = \dfrac{1}{R}, \qquad i = G\, u$	Siemens: 1 S = 1 Ω^{-1}.
Joule's law Box 13, eq. (7)	$p = u\, i = R\, i^2 = \dfrac{u^2}{R}$	Power turned into heat by R (W). It is $R\, i^2$, never $i\, R^2$.
Heat released†	$E_J = R\, i^2\, \Delta t$	Constant current i for a time Δt.

Typical resistances (Tab. 6): lab lead, a few tens of mΩ; ammeter, 10 mΩ to a few Ω; lab resistor, 1 Ω to 1 MΩ; dry skin, a few kΩ to tens of kΩ; voltmeter input, 1 to 10 MΩ.

2.C Pouillet's law (pp. 21–22)

Pouillet's law Box 14	$R = \rho\, \dfrac{L}{S}$	ρ resistivity (Ω m), L length (m), S cross-section (m^2). Circular wire: $S = \pi d^2/4$.
Conductivity Box 15	$\sigma = \dfrac{1}{\rho}, \qquad R = \dfrac{L}{\sigma\, S}$	S m^{-1}. Good conductor: small ρ, large σ.
Scaling†	$R \propto \dfrac{L}{d^2}$	Doubling the diameter divides R by 4.

Material (Tab. 7)	Silver	Copper	Aluminium	Iron	Graphite
ρ (Ω m)	1.6×10^{-8}	1.7×10^{-8}	2.8×10^{-8}	1.0×10^{-7}	$\sim 3 \times 10^{-5}$
σ (S m^{-1})	6.3×10^{7}	5.9×10^{7}	3.6×10^{7}	1.0×10^{7}	$\sim 3 \times 10^{4}$

For most metals ρ increases with temperature. Glass: ρ very large.

3.A Kirchhoff's laws and their consequences (pp. 22–29)

Node law Box 16	$\sum i_{\text{in}} = \sum i_{\text{out}}$	Conservation of charge: nothing piles up at a node.
Mesh law Box 16	$\sum_{\text{mesh}} \varepsilon_k\, u_k = 0$	Conservation of energy. $\varepsilon_k = +1$ if the arrow of u_k points along the direction of travel, -1 otherwise.
Resistors in series Box 17, eq. (8)	$R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$	Larger than the largest R_k; n identical: nR.

Resistors in parallel Box 18, eq. (9)	$\dfrac{1}{R_{\text{eq}}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots + \dfrac{1}{R_n}$	$G_{\text{eq}} = \sum G_k$. Smaller than the smallest R_k; n identical: R/n.
Two in parallel eq. (10)	$R_{\text{eq}} = \dfrac{R_1 R_2}{R_1 + R_2}$	Product over sum.
Voltage divider Box 19, eq. (12)	$u_1 = \dfrac{R_1}{R_1 + R_2}\, u, \qquad u_2 = \dfrac{R_2}{R_1 + R_2}\, u$	Same current in R_1 and R_2 (nothing drawn from the midpoint). $u_1/u_2 = R_1/R_2$. With a load R_L across R_2: use $R_2 \parallel R_L$.
Current divider†	$i_1 = \dfrac{R_2}{R_1 + R_2}\, i = \dfrac{G_1}{G_1 + G_2}\, i$	Two resistors in parallel sharing i (crossed indices).
Millman's theorem Box 20, eq. (17)	$V_O = \dfrac{\sum_k V_k/R_k}{\sum_k 1/R_k}$	Node O joined only through resistors R_k to potentials V_k. Source E_k in the branch: $V_k = \pm E_k$; branch to ground: $V_k = 0$ but $1/R_k$ stays below.

3.B Thévenin's and Norton's theorems (pp. 29–33)

Thévenin model Fig. 25	$u = E_{\text{Th}} - R_{\text{Th}}\, i$	Ideal source E_{Th} in series with R_{Th}. Generator convention: i leaves by A, $u = V_A - V_B$.
Norton model Fig. 26	$i = I_{\text{No}} - \dfrac{u}{R_{\text{No}}}$	Ideal current source I_{No} in parallel with R_{No}.
Equivalence of the models Box 21, eq. (18)	$R_{\text{Th}} = R_{\text{No}} \equiv R, \qquad E_{\text{Th}} = R\, I_{\text{No}}$	E_{Th}: open-circuit voltage ($i = 0$). I_{No}: short-circuit current ($u = 0$).
Thévenin's theorem Box 22	$E_{\text{Th}} = u_{AB}\big\vert_{i=0}, \qquad R_{\text{Th}} = R_{AB}\big\vert_{\text{sources off}}$	Valid for ideal sources and ohmic conductors only (no lamp, motor, diode).
Norton's theorem Box 23	$I_{\text{No}} = i_{A\to B}\big\vert_{u=0}, \qquad R_{\text{No}} = R_{\text{Th}}$	I_{No}: current in a wire joining A and B, sources on.
Switching a source off p. 32	voltage source → wire ($u = 0$) current source → open circuit ($i = 0$)	The resistors stay in place.
Source transformation†	E in series with $R \;\Leftrightarrow\; \dfrac{E}{R}$ in parallel with R	Thévenin models in series add E and R; Norton models in parallel add I and G.
Superposition†	$x = x_1 + x_2 + \ldots$	Linear circuit: the response to several sources is the sum of the responses to each source alone, the others switched off.
Thévenin model of a divider p. 33	$R_{\text{Th}} = \dfrac{R_1 R_2}{R_1 + R_2}, \qquad E_{\text{Th}} = \dfrac{R_2}{R_1 + R_2}\, E$	E in series with R_1, R_2 across A–B. Then $I_{\text{No}} = E/R_1$.

3.C Operating point (pp. 34–38)

Characteristics Figs. 28, 29	source: $u = E_{\text{Th}} - R_{\text{Th}}\, i$ load: $u = R_c\, i$	Operating point $(u^\star, i^\star)$: the unique pair on both; graphically their intersection.
Operating point Box 24, eq. (21)	$i^\star = \dfrac{E_{\text{Th}}}{R_{\text{Th}} + R_c}, \qquad u^\star = \dfrac{R_c}{R_{\text{Th}} + R_c}\, E_{\text{Th}}$	$R_c \to 0$: $u^\star \to 0$, $i^\star \to E_{\text{Th}}/R_{\text{Th}}$. $R_c \to \infty$: $i^\star \to 0$, $u^\star \to E_{\text{Th}}$.
Power received by the load eqs. (19), (20)	$P_c = u^\star\, i^\star = R_c\, i^{\star 2} = \dfrac{R_c\, E_{\text{Th}}^2}{(R_{\text{Th}} + R_c)^2}$	Equal to the power delivered by the real source at A–B.
Power balance eqs. (22), (23)	$E_{\text{Th}}\, i^\star = R_{\text{Th}}\, i^{\star 2} + R_c\, i^{\star 2}$	Produced by the ideal source = lost in R_{Th} + received by the load.
Efficiency eq. (24)	$\eta = \dfrac{P_c}{E_{\text{Th}}\, i^\star} = \dfrac{R_c}{R_{\text{Th}} + R_c}$	0 on short circuit, close to 1 for $R_c \gg R_{\text{Th}}$, $\frac{1}{2}$ at $R_c = R_{\text{Th}}$.
Maximum power (matching) eqs. (25), (26)	$R_c = R_{\text{Th}} \implies P_c^{\max} = \dfrac{E_{\text{Th}}^2}{4R_{\text{Th}}}$	$\mathrm{d}P_c/\mathrm{d}R_c = E_{\text{Th}}^2 (R_{\text{Th}} - R_c)/(R_{\text{Th}} + R_c)^3$. Efficiency only 50 % at matching: maximum power ≠ maximum efficiency.

Misprints and slips in the Electricity notes

- Eq. (17): the denominator of Millman's formula is $1/R_1 + 1/R_2 + \cdots + 1/R_n$.
- Eqs. (15)–(16): the left-hand side is u_2 (the voltage across R_2), not u_1; "0.9901 E" means $0.9901\,u$.
- Eq. (24): P_{Th} there means $E_{\text{Th}} i^\star$ (produced by the ideal source), not the delivered power of eq. (19).
- Tabs. 3–4: the five ratios u_k/i_k average 50.04 Ω (printed 50.17 Ω); the regression gives $\hat{R} = 49.99\ \Omega \approx 50.0\ \Omega$; with these residuals the uncertainty formula gives $u(\hat{R}) = 0.08\ \Omega$ (printed 1.001 Ω).

Part II
Thermodynamics: description of a system

Notes Thermo-S1 Part 1 (23 pages). Closed systems only: n is constant.

1.A System and state variables (pp. 2–5)

Kinds of system Boxes 1–3	closed: no matter exchanged isolated: neither matter nor energy	Open otherwise. Isolated ⇒ closed, but not the converse.
State variables of a gas Tab. 1	p (Pa), V (m^3), T (K), n (mol)	Dimensions: $\mathrm{M\,L^{-1}T^{-2}}$, $\mathrm{L^3}$, Θ, N. Defined only at equilibrium (Box 4).
Extensive quantity Box 5, eq. (1)	$X(2 \times \text{system}) = 2\,X$	Proportional to the size: m, V, n, E.
Intensive quantity Box 6	$X(2 \times \text{system}) = X$	p, T, constants such as R. Ratio of two extensive quantities: intensive (V/n).
Homogeneity Box 7	intensive = intensive, extensive = extensive	A check besides dimensions: $T = pV/(nR)$ passes, $T = nR$ fails.

Composition quantities (Tab. 2)

Mole fraction Tab. 2	$x_i = \dfrac{n_i}{n_{\text{tot}}}, \quad \sum_i x_i = 1$	No unit.
Molar density Tab. 2	$C = \dfrac{n}{V}$	$\mathrm{mol\,m^{-3}}$.
Molar volume Tab. 2	$V_m = \dfrac{V}{n}$	$\mathrm{m^3\,mol^{-1}}$ (inverse of C).
Mass density Tab. 2	$\rho = \dfrac{m}{V}$	$\mathrm{kg\,m^{-3}}$ (also written μ).
Molar mass Tab. 2	$M = \dfrac{m}{n}$	$\mathrm{g\,mol^{-1}}$.
Mass concentration Tab. 2	$c_m = \dfrac{m_{\text{solute}}}{V_{\text{solution}}}$	$\mathrm{g\,L^{-1}}$ (also γ).
Molar concentration Tab. 2	$c = \dfrac{n_{\text{solute}}}{V_{\text{solution}}}$	$\mathrm{mol\,L^{-1}}$ (also [X]). $c_m = c\,M$.
Number of entities†	$N = n\,N_A$	$N_A = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$.

1.B Postulates of thermodynamics (pp. 5–7)

Thermodynamic equilibrium eqs. (2), (3)	$T_{\text{system}} = T_{\text{ext}}, \quad p_{\text{system}} = p_{\text{ext}}$	Thermal and mechanical equilibrium: uniform and steady state variables.
Postulate Box 8	a few state variables describe an equilibrium state	For a closed system: p, V, T.
Pressure†	$p = \dfrac{F}{S}$	Pa = $\mathrm{N\,m^{-2}}$; force F normal to the area S.
Piston carrying a mass Fig. 5	$p = p_0 + \dfrac{m\,g}{S}$	Mechanical equilibrium of a vertical piston of mass m and area S under the outside pressure p_0.

Hydrostatic pressure†	$p = p_0 + \rho\, g\, h$	Depth h in a liquid of density ρ.

1.C Equations of state and state functions (pp. 7–10)

Equation of state Box 9, eq. (4)	$\psi(p, V, T) = 0 \iff V = V(T, p)$	Valid only at equilibrium; characterises the material.
Differential of a state function eq. (5)	$\mathrm{d}V = \left(\frac{\partial V}{\partial T}\right)_p \mathrm{d}T + \left(\frac{\partial V}{\partial p}\right)_T \mathrm{d}p$	Each partial derivative is the slope of an isobar $V(T)$ or an isotherm $V(p)$ (Fig. 7).
Cyclic rule eq. (11)	$\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T = -1$	For any relation $f(p, V, T) = 0$.
Inverse derivative†	$\left(\frac{\partial V}{\partial p}\right)_T = \frac{1}{(\partial p/\partial V)_T}$	Same variable held fixed.

2.A Condensed phases: χ_T and α (pp. 11–13)

Isothermal compressibility Box 10, eq. (7)	$\chi_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T$	Pa^{-1}, positive. Condensed phases: $\chi_T \approx 0$.
Isobaric expansivity Box 11	$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$	K^{-1}; nearly constant over a limited range of T. Also $\alpha = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_p$.
Relative variations†	$\frac{\mathrm{d}V}{V} = \alpha\, \mathrm{d}T - \chi_T\, \mathrm{d}p$	Small changes: $\frac{\Delta V}{V} \approx \alpha\, \Delta T - \chi_T\, \Delta p$, and $\frac{\Delta \rho}{\rho} = -\frac{\Delta V}{V}$.
Equation of state of a condensed phase Box 13	$V = V_0\, e^{\alpha(T-T_0)-\chi_T(p-p_0)}$ $\approx V_0\,[1 + \alpha(T - T_0) - \chi_T(p - p_0)]$	α, χ_T constant; integrate $\mathrm{d}V/V$. Limit $\alpha = \chi_T = 0$: $V = V_0$.
Heating at constant volume†	$\left(\frac{\partial p}{\partial T}\right)_V = \frac{\alpha}{\chi_T} \qquad \Delta p \approx \frac{\alpha}{\chi_T}\, \Delta T$	Liquid sealed in a rigid container ($\mathrm{d}V = 0$).

Orders of magnitude†: water $\alpha \approx 2\times10^{-4}\ \mathrm{K}^{-1}$, $\chi_T \approx 5\times10^{-10}\ \mathrm{Pa}^{-1}$; ideal gas at 1 bar and 300 K $\chi_T = 10^{-5}\ \mathrm{Pa}^{-1}$, $\alpha = 3.3\times10^{-3}\ \mathrm{K}^{-1}$.

2.B Gases: β (pp. 13–14)

Piezothermic coefficient Box 12	$\beta = \frac{1}{p}\left(\frac{\partial p}{\partial T}\right)_V$	K^{-1}: relative rise of the pressure with T at constant volume.

2.C Principle and method (pp. 14–15)

Link between the three eq. (11)	$\beta = \frac{\alpha}{p\,\chi_T} \iff \alpha = p\,\beta\,\chi_T$	From the cyclic rule applied to $f(T, V, p) = 0$.
Method Fig. 8	equation of state $\underset{\text{integrate}}{\overset{\text{differentiate}}{\rightleftharpoons}}$ $\alpha,\ \chi_T,\ \beta$	Coefficients measured by experiment give the equation of state, and conversely.

3.A Low-density gases: the historical laws (pp. 15–17)

Boyle–Mariotte (1662) eq. (12)	$pV = \text{const}$	$(n, T \text{ fixed})$	$p_1V_1 = p_2V_2$. Gives $\chi_T = 1/p$ (Box 14).
Charles (1787) eq. (13)	$\frac{V}{T} = \text{const}$	$(n, p \text{ fixed})$	$V_1/T_1 = V_2/T_2$. Gives $\alpha = 1/T$ (Box 15).
Gay-Lussac (1802) eq. (14)	$\frac{p}{T} = \text{const}$	$(n, V \text{ fixed})$	$p_1/T_1 = p_2/T_2$. Gives $\beta = 1/T$ (Box 16).
Avogadro (1811) p. 15	$\frac{V}{n} = \text{const}$	$(p, T \text{ fixed})$	Same p, T and V ⇒ same amount of gas.

3.B The ideal gas (pp. 17–20)

Ideal gas law (Clapeyron) Box 17	$pV = nRT$	p in Pa, V in m^3, T in K, $R = 8.314\ \mathrm{J\,mol^{-1}\,K^{-1}}$. Limit of every gas when $n/V \to 0$ (no interactions).
Derivation Box 18	$\dfrac{pV}{T} = \text{const} = nR$	Combine Boyle–Mariotte, Charles and Gay-Lussac; Avogadro gives the factor n.
Two states, n fixed†	$\dfrac{p_1V_1}{T_1} = \dfrac{p_2V_2}{T_2}$	Identify first which variables are fixed (table below).
Other forms†	$pV_m = RT, \quad \rho = \dfrac{pM}{RT}, \quad C = \dfrac{p}{RT}$	M in $\mathrm{kg\,mol^{-1}}$ for ρ in $\mathrm{kg\,m^{-3}}$.
Per molecule†	$pV = N\,k_B\,T$	$k_B = R/N_A = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$.
Coefficients of the ideal gas Boxes 14–16	$\chi_T = \dfrac{1}{p}, \quad \alpha = \dfrac{1}{T}, \quad \beta = \dfrac{1}{T}$	Check: $\alpha/(p\,\chi_T) = 1/T = \beta$.
Molar volume†	$V_m = \dfrac{RT}{p}$	$22.4\ \mathrm{L\,mol^{-1}}$ at 0 °C and 1 atm; $24.8\ \mathrm{L\,mol^{-1}}$ at 25 °C and 1 bar.
Mixture (Dalton)†	$p_i = x_i\,p, \quad p = \sum_i p_i$	Partial pressures. Mean molar mass $M = \sum_i x_i M_i$; air $\approx 29\ \mathrm{g\,mol^{-1}}$.

System (Tab. 5)	n	T	p	V	Law to use
Weather balloon	fixed	free	free ($= p_{\mathrm{atm}}$)	free	$pV/T = \text{const}$
Tyre	fixed	free	free	fixed	$p/T = \text{const}$
Cartesian diver	fixed	fixed	free ($\propto$ depth)	free	$pV = \text{const}$
Hot-air balloon	free	free	fixed ($= p_{\mathrm{atm}}$)	fixed	$nT = \text{const}$

3.C Real gases (pp. 20–23)

Compressibility factor eq. (15)	$Z = \dfrac{pV}{nRT} = \dfrac{pV_m}{RT}$	$Z = 1$: ideal gas. $Z < 1$: attractions dominate. $Z > 1$: the volume of the molecules dominates. $p \to 0$: $Z \to 1$ (Amagat diagram: Z against p).
Boyle temperature p. 22	$T = T_B$: $Z \approx 1$ over a wide range of p	Below T_B, Z first dips below 1; well above, $Z > 1$.
Van der Waals eq. (16)	$\left(p + a\,\dfrac{n^2}{V^2}\right)(V - nb) = nRT$	a ($\mathrm{Pa\,m^6\,mol^{-2}}$): attractions; b ($\mathrm{m^3\,mol^{-1}}$): co-volume. $a, b \to 0$: ideal gas.
Van der Waals, other forms†	$\left(p + \dfrac{a}{V_m^2}\right)(V_m - b) = RT$ $p = \dfrac{nRT}{V - nb} - a\,\dfrac{n^2}{V^2}$	Per mole ($V_m = V/n$), and solved for p.
Internal pressure, co-volume p. 23	$\Pi = a\left(\dfrac{n}{V}\right)^2, \quad V_{\mathrm{excl}} = nb$	nb: about four times the volume of the molecules.
Co-volume gas ($a = 0$)†	$p\,(V - nb) = nRT$	Repulsions only.
Virial expansion†	$Z = 1 + \dfrac{B(T)}{V_m} + \dfrac{C(T)}{V_m^2} + \ldots$	Van der Waals: $B = b - \frac{a}{RT}$, zero at $T_B = \frac{a}{Rb}$.
Van der Waals critical point†	$T_c = \dfrac{8a}{27Rb}, \quad p_c = \dfrac{a}{27b^2}, \quad V_{m,c} = 3b$	Reproduces the T_c column of Tab. 7.

Gas (Tab. 7)	He	H_2	N_2	CO_2	H_2O
a ($\mathrm{Pa\,m^6\,mol^{-2}}$)	3.46×10^{-3}	2.45×10^{-2}	0.137	0.364	0.554
b ($10^{-5}\ \mathrm{m^3\,mol^{-1}}$)	2.38	2.67	3.87	4.27	3.05
T_c (K)	5.2	33.2	126	304	647

Other equations of state (Tab. 6, for reference): Clausius $\left(p+\frac{a}{T(V_m+c)^2}\right)(V_m-b)=RT$; Berthelot $\left(p+\frac{a}{TV_m^2}\right)(V_m-b)=RT$; Dieterici $p\,(V_m-b)\,e^{a/(V_m RT)}=RT$; Soave–Redlich–Kwong $p=\frac{RT}{V_m-b}-\frac{a\,\alpha(T)}{V_m(V_m+b)}$; Peng–Robinson $p=\frac{RT}{V_m-b}-\frac{a\,\alpha(T)}{V_m(V_m+b)+b(V_m-b)}$.

Part III†
Work, heat and heat engines (beyond Part 1)

Not in the Part 1 notes: the standard results for ideal-gas cycles. W and Q are the energies **received** by the gas (>0 if received); temperatures in kelvin.

III.1 First law and the ideal gas

Work of the pressure forces†

$$\delta W=-p_{\text{ext}}\,\mathrm{d}V$$
$$\text{reversible: } W=-\int_{V_1}^{V_2}p\,\mathrm{d}V$$

$W>0$ in a compression. Area under the curve in the (p,V) diagram.

First law†

$$\Delta U=W+Q$$

Closed system at rest. U is a state function; W and Q depend on the path. Over a cycle $\Delta U=0$.

Internal energy of an ideal gas†

$$\Delta U=n\,C_{V,m}\,\Delta T$$

Depends on T only. $C_{V,m}=\frac{3}{2}R$ (monatomic), $\frac{5}{2}R$ (diatomic near 300 K).

Enthalpy†

$$H=U+pV,\qquad \Delta H=n\,C_{p,m}\,\Delta T$$

At constant pressure: $Q=\Delta H$.

Mayer's relation, γ†

$$C_{p,m}-C_{V,m}=R,\qquad \gamma=\frac{C_{p,m}}{C_{V,m}}$$

$\gamma=5/3$ (monatomic), 7/5 (diatomic).

Reversible process	Constant	W (received)	Q (received)	ΔU
Isochoric	V	0	$n\,C_{V,m}\,\Delta T$	$n\,C_{V,m}\,\Delta T$
Isobaric	p	$-p\,\Delta V=-nR\,\Delta T$	$n\,C_{p,m}\,\Delta T$	$n\,C_{V,m}\,\Delta T$
Isothermal	T	$-nRT\ln\frac{V_2}{V_1}$	$nRT\ln\frac{V_2}{V_1}$	0
Adiabatic	$Q=0$	$\frac{p_2V_2-p_1V_1}{\gamma-1}$	0	$n\,C_{V,m}\,\Delta T$

Laplace's law†

$$pV^{\gamma}=\text{const},\qquad TV^{\gamma-1}=\text{const}$$
$$T^{\gamma}p^{1-\gamma}=\text{const}$$

Reversible adiabatic process of an ideal gas with constant γ.

III.2 Second law and heat engines

Second law (entropy)†

$$\Delta S=S_{\text{exchanged}}+S_{\text{created}},\qquad S_{\text{created}}\geq 0$$
$$S_{\text{exchanged}}=\sum_k\frac{Q_k}{T_k}$$

T_k: temperature of the source supplying Q_k. $S_{\text{created}}=0$ only for a reversible process.

Entropy of an ideal gas†

$$\Delta S=n\,C_{V,m}\ln\frac{T_2}{T_1}+nR\ln\frac{V_2}{V_1}$$

$\mathrm{J\,K^{-1}}$.

Heat engine (two sources)†

$$W+Q_H+Q_C=0,\qquad \eta=\frac{-W}{Q_H}=1+\frac{Q_C}{Q_H}$$

$Q_H>0$ taken from the hot source, $Q_C<0$ given to the cold one, $W<0$ delivered.

Carnot (maximum) efficiency†

$$\eta\leq\eta_C=1-\frac{T_C}{T_H}$$

Equality for a reversible engine, where $\frac{Q_H}{T_H}+\frac{Q_C}{T_C}=0$.

Refrigerator and heat pump†

$$e_{\text{fridge}}=\frac{Q_C}{W}\leq\frac{T_C}{T_H-T_C}$$
$$e_{\text{pump}}=\frac{-Q_H}{W}\leq\frac{T_H}{T_H-T_C}$$

Coefficients of performance; they can exceed 1.

Carnot cycle of an ideal gas†	$Q_H = nRT_H \ln\dfrac{V_2}{V_1}, \qquad Q_C = -nRT_C \ln\dfrac{V_3}{V_4}$ $\dfrac{V_2}{V_1} = \dfrac{V_3}{V_4}$	Isotherm T_H ($V_1 \to V_2$), adiabat ($\to V_3 = V_{\max}$), isotherm T_C ($\to V_4$), adiabat back to $V_1 = V_{\min}$. Here $Q_H \neq nRT_H \ln(V_{\max}/V_{\min})$.
Stirling cycle†	$Q_{\text{in}} = nRT_H \ln\dfrac{V_{\max}}{V_{\min}}$ $W = -nR\,(T_H - T_C) \ln\dfrac{V_{\max}}{V_{\min}}$	Two isotherms and two isochores. With an ideal regenerator, $\eta = 1 - T_C/T_H$. Without one, add $n\,C_{V,m}(T_H - T_C)$ to Q_{in}.

Summary
Units of all quantities

Electricity

Charge	q	coulomb, $\mathrm{C} = \mathrm{A\,s}$
Current	i	ampere, A (SI base unit)
Potential, voltage	$V,\ u,\ e$	volt, $\mathrm{V} = \mathrm{J\,C^{-1}}$
Resistance	R	ohm, $\Omega = \mathrm{V\,A^{-1}}$
Conductance	G	siemens, $\mathrm{S} = \Omega^{-1}$
Resistivity	ρ	$\Omega\,\mathrm{m}$
Conductivity	σ	$\mathrm{S\,m^{-1}}$
Power	$p,\ P$	watt, $\mathrm{W} = \mathrm{V\,A}$
Energy	E	joule, $\mathrm{J} = \mathrm{W\,s}$
Carrier density	n	$\mathrm{m^{-3}}$
Drift velocity	v	$\mathrm{m\,s^{-1}}$
Current density	j	$\mathrm{A\,m^{-2}}$
Efficiency	η	no unit

Thermodynamics

Pressure	p	pascal, $\mathrm{Pa} = \mathrm{N\,m^{-2}}$
Volume	V	$\mathrm{m^3}$
Temperature	T	kelvin, K
Amount of matter	n	mole, mol
Molar mass	M	$\mathrm{kg\,mol^{-1}}$ (or $\mathrm{g\,mol^{-1}}$)
Molar volume	V_m	$\mathrm{m^3\,mol^{-1}}$
Density	ρ	$\mathrm{kg\,m^{-3}}$
Coefficients	$\alpha,\ \beta;\ \chi_T$	$\mathrm{K^{-1}}$; $\mathrm{Pa^{-1}}$
Gas constant	R	$\mathrm{J\,mol^{-1}\,K^{-1}}$
Van der Waals a	a	$\mathrm{Pa\,m^6\,mol^{-2}}$
Van der Waals b	b	$\mathrm{m^3\,mol^{-1}}$
Energies	$W,\ Q,\ U,\ H$	joule, J
Entropy	S	$\mathrm{J\,K^{-1}}$
Heat capacities	$C_{V,m},\ C_{p,m}$	$\mathrm{J\,mol^{-1}\,K^{-1}}$

Same letter, different quantity

p: power (W) in electricity, pressure (Pa) in thermodynamics. ρ: resistivity ($\Omega\,\mathrm{m}$) or density ($\mathrm{kg\,m^{-3}}$). n: carrier density ($\mathrm{m^{-3}}$) or amount of substance (mol). α: expansivity here, never a resistance coefficient. E: energy (J) or electromotive force E_{Th} (V).